COURSE SYLLABUS
CSC14119 – Introduction to Data Science

1. GENERAL INFORMATION

Course name:	Introduction to Data Science
Course name (in Vietnamese):	Nhập môn Khoa học dữ liệu
Course ID:	CSC14119
Knowledge block:	Specialized knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	
Instructors:	Lê Ngọc Thành, PhD. Nguyễn Ngọc Thảo, PhD.

2. COURSE DESCRIPTION

This course provides an introduction to the Data Science (DS) pipeline, offering a structured overview of each phase and its role in the analytical process. Students will gain both conceptual understanding and practical exposure to the tools and techniques commonly applied at each stage.

The course begins with data acquisition, focusing on collecting relevant datasets from web sources and storing them in suitable repositories. Students will then examine methods for preparing and transforming raw data into structured formats optimized for feature extraction and analysis. Building on this foundation, they will develop decision-making models to generate actionable insights and evaluate their effectiveness. The course concludes with presenting results through effective data visualization techniques.

By the end of the course, students will have developed a clear understanding of the fundamental stages of the Data Science process, equipping them with the skills and knowledge to pursue advanced study or applied research in the field.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Explain the fundamental concepts and terminologies of Data science	
G2	Apply suitable techniques to each step of the Data science process	
G3	Manipulate tools and libraries commonly used in Data science	
G4	Promote personal aptitudes of logical thinking and communication	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Define fundamental concepts and differentiate terminologies of Data science	I
G1.2	Understand textbooks and tutorials about Data science	U
G2.1	Interpret the web crawlers and their related issues	I
G2.2	List common data repositories for Data science and describe their functionalities	I
G2.3	Explore the data with Exploratory Data Analysis (EDA)	T
G2.4	Demonstrate the feature engineering process on different types of data	T
G2.5	Construct regression models from data and evaluate them with objective metrics	T
G3.1	Use popular tools and libraries to crawl data from web pages and extract required information from crawled data	T/U
G3.2	Employ some data storage techniques to store the data after crawling	T/U
G3.3	Apply Python Pandas and scikit-learn for data exploration	T/U
G3.4	Apply tools and libraries to extract features from different types of data	T/U
G3.5	Apply scikit-learn to build learning models from data and evaluate the models	T/U
G4.1	Develop practical personal communication skills, both oral and writing	U
G4.2	Develop the personal aptitudes of logical thinking	U

5. TEACHING PLAN

Week	Topic	Course outcomes	Teaching/Learning Activities (S: Student, T: Teacher)	Assessment
1	Introduction to Data science	G1.1-2	**In class** • Announce the syllabus and regulations (T) • Lecturing (T)	
2	Data collection	G1.1-2 G2.1 G3.1	**Before class:** • Read the lecture slide (S) • Prepare the tools: Selenium (web scraper) and Beautiful Soup (HTML parser) **In class** • Lecturing and Demonstration (T) • Complete case studies about crawling data (S)	
3	Data management	G1.1-2 G2.2 G3.2	**Before class:** • Read the lecture slide (S) • Prepare the tools: Pandas (S) **In class** • Lecturing and Demonstration (T) • Complete case studies about handling data (S)	
4-5	Exploratory data analysis	G1.1-2 G2.3 G3.3	**Before class:** • Read the lecture slide (S) • Prepare the tools: scikit-learn and matplotlib (S) **In class** • Lecturing and Demonstration (T) • Complete case studies about basic EDA (S)	
6	**Midterm exam (optional)**	G2.1-3		
7-8	Feature engineering	G2.4 G3.4	**Before class:** • Read the lecture slide (S) • Prepare the tools: scikit-learn, scikit-image, Hugging Face BERT, and node2vec **In class** • Lecturing and Demonstration (T) • Complete case studies about feature extraction (S)	

Week	Topic	Course outcomes	Teaching/Learning Activities (S: Student, T: Teacher)	Assessment
9-10	Data modeling	G2.5 G3.5	**Before class:** • Read the lecture slide (S) **In class** • Lecturing and Demonstration (T) • Complete case studies about model construction and evaluation (S)	
11	Review			

** Please note that the lecturer reserves the right to modify the sequence of topics in response to the class's circumstances.*

6. LABORATORY WORK PLAN

The teaching assistant and lab instructors are responsible for

- Consolidating students' problem-solving and technical skills on frameworks and tools, and
- Organizing Q&A sessions for lab work (on demand), and
- Giving, correcting, and grading assignments.

Students will not have weekly classes for laboratory work. Instead, they will contact TA or lab instructors when necessary.

Week	Topic	Course outcomes	Teaching/Learning Activities	Assessments
1			Self-study: Python programming	
2	Data collection	G3.1	Experience the tools (S) • Selenium (web scraper) • Beautiful Soup (HTML parser)	
3	Data management	G3.2	Project: Announcement and QA (T) Experience Pandas (S)	P1: Course project
4-5	Exploratory data analysis	G3.3	Experience the tools (S): • scikit-learn for EDA and basic visualization • matplotlib	
6	*Students have theoretical midterm.*			

7-8	Feature engineering	G3.4	Experience the tools (S): • scikit-learn TF-IDF and Hugging Face BERT for text data • scikit-image for image data • node2vec for graph data	
9-10	Data modeling	G3.5	Experience scikit-learn for models and evaluation (S)	
11	Review			

7. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	**Group assignments: 3-4 members per group**			**50**
A11	Course project (P1)	The project includes *a Python program* containing solutions for the given problems and *a written report*.	G1.1-2 G2.1-5 G3.1-5 G4.1-2	20
A12	Labs	Take-home coding assignments	G1.1-2 G2.1-5 G3.1-5 G4.1-2	30
A2	**Examinations: individual work**			**50-60**
A21	Midterm exam/Activity Class	*For Midterm: 60 minutes, open-book, in-class written exam* They are on any topics in any lecture covered and any reading material assigned up to the time the exam is administered	G1.1-2 G2.1-3 G3.1-3	10-20
A22	Final exam	*90 minutes, open-book, in-class written exam* They are on any topics in any lecture covered and any reading material assigned up to the time the exam is administered	G1.1-2 G2.1-5 G3.1-5	40

8. RESOURCES

Textbooks

- Alan Agresti, Christine A. Franklin, Bernhard Klingenberg. **Statistics: The Art and Science of Learning from Data**. Global Edition 4th Edition. Pearson, 2018.
- McKinney, Wes. **Python for data analysis: Data wrangling with Pandas, NumPy, and IPython**. Second edition. O'Reilly Media, Inc., 2012.

Other materials

- Han, J., Kamber, M., & Pei, J. **Data mining: Concepts and techniques**. Third edition. Morgan Kaufmann Publishers, 2012.

Multimedia materials

- The list of materials will be updated every term following the evolution of Data science research.
- Learning from Data, http://work.caltech.edu/telecourse.html
- CS109 Data Science, http://cs109.github.io/2015/
- The Analytics Edge, https://www.edx.org/course/analytics-edge-mitx-15-071x-2

Tools

- Python 3, Jupyter notebook with Anaconda, Pandas, and scikit-learn
- Web crawlers, HTML parser

9. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.